Machine Learning Project Report

Group-M

Group members:

- Lucrezia Salerno 70098
- Guilherme Morgado 56857
- Miguel Teixeira 56529
- Isaac Carvalho 57045
- Amal Bouilla 70973
- Guilherme Carvalho 70364

Table of contents:

1. Introduction

With teams spending billions of euros on player acquisitions annually, the football transfer market has become more complex and expensive. Even with the market's enormous financial scale, a sizable percentage of transactions fall short of the intended outcomes, particularly for high-value signings. These inefficiencies can hurt a club's long-term competitive success in addition to being costly.

Due to their tight budgets and inability to make needless purchases, mid-tier teams are particularly vulnerable to this problem. Because these institutions, unlike elite clubs, have to maximize the value of each transfer move, mistakes have a much greater impact.

While essential, traditional scouting techniques are usually subjective and have a limited scope. Due to their reliance on observation and experience, scouts may be biased and lose important statistical trends. Because of this, teams often overvalue players who are very well-known while ignoring others who might perform better for a lower cost.

In this context, the objective of this project is to develop a Machine Learning model capable of predicting the fair market value of professional football players. By leveraging data-driven techniques, the model aims to support more informed transfer decisions and identify players who may be undervalued in the market.

Lastly, our approach seeks to bridge the gap between decision-making and raw data, providing clubs with a more objective and scalable framework for player evaluation.

2. Business Understanding

A sports analytics system designed for mid-tier football teams with budgets between €50 million and €200 million is the setting for this project. The core problem addressed is the inefficiency of player valuation in the transfer market. Clubs often rely on subjective assessments and incomplete information, which can lead to overpaying or failing to identify valuable opportunities.

The proposed solution consists of a Machine Learning model that predicts player market value based on performance data, contextual information, and career-related features such as goals, assists, minutes played, age, league quality, and contract conditions. By comparing predicted values with actual market prices, the system enables the identification of undervalued players who represent opportunities for clubs to acquire talent at a lower cost.

From a business perspective, success is measured by the model's ability to improve decision-making, reduce financial risk, and optimize resource allocation. Player values are influenced not only by performance but also by external factors such as league reputation, contract duration, and market trends. By incorporating these contextual variables, the solution becomes more aligned with real-world valuation mechanisms, combining quantitative analysis with domain knowledge to create a holistic framework for player valuation.

3. Data and Methodology

The dataset is constructed by merging five Transfermarkt sources: player profiles, performance statistics, market value history, injury records, and team information. After integration, the final dataset contains 33,557 players with 40 engineered features. Performance data is aggregated across seasons from 2010 onwards using a common player identifier.

Data preprocessing includes median imputation for numeric columns, "Unknown" replacement for categorical variables, and conversion of date fields into numeric durations. An important data quality issue was identified: out of 1,350 players with zero minutes played, 687 had non-zero performance statistics, which is logically impossible. All performance metrics for these players were reset to zero. Player identifiers, names, and raw date strings were removed to prevent data leakage.

Feature engineering generates 40 features organized into several groups. Age-related features include age, age squared to capture the non-linear peak-value curve, and a career phase variable (Youth, Prime, or Veteran) assigned using position-specific thresholds — for example, attackers enter their prime at 22 and become veterans at 28, while goalkeepers peak later between 23 and 32. Contract features include years remaining under contract and a binary free agent flag. League quality is encoded as a five-tier ordinal scale, where elite leagues receive the highest tier and youth competitions the lowest. Normalized performance metrics are computed on a per-90-minute basis, including goals, assists, goal contributions, and cards per 90 minutes, along with a composite net performance score and its position-normalized z-score. Interaction terms capture complex relationships: goal contributions normalized multiplied by league quality reflects that the same output is worth more in a top-five league, while age multiplied by position encodes that a 30-year-old goalkeeper is valued differently from a 30-year-old striker. A preliminary PCA on the 16 numeric features confirms meaningful redundancy, with 7 components explaining 80% and 9 explaining 90% of variance.

The data is split 80/20 into training (26,845) and test (6,712) sets. A ColumnTransformer applies StandardScaler to numeric features and OneHotEncoder to categorical features, embedded inside scikit-learn Pipeline objects to prevent data leakage.

Seven models are trained: Linear Regression as a baseline; Ridge (best alpha: 10.0), Lasso (best alpha: 0.01), and ElasticNet (best alpha: 0.01, L1 ratio: 0.7) tuned via cross-validation; a Random Forest with 100 trees and max depth 15; a tuned Random Forest optimized via RandomizedSearchCV (200 trees, max depth 25); and a Neural Network with three hidden layers (256, 128, 64 neurons), ReLU activation, Adam optimizer, and log-transformed target to address the extreme skew of market values ranging from €10,000 to €200,000,000.

4. Results and Discussion

The results reveal a clear separation between model families. The four linear models cluster around $R^2 \approx 0.52$ and MAE ≈ €1.28 million. The non-linear models achieve substantially better results: the Random Forest reaches $R^2 = 0.7365$, RMSE = €3,071,037, and MAE = €773,568; the tuned Random Forest achieves $R^2 = 0.7316$, RMSE = €3,099,503, and MAE = €765,753; and the Neural Network reaches $R^2 = 0.7137$, RMSE = €3,200,966, and MAE = €800,663. This gap confirms that market value is driven by complex non-linear interactions that linear models cannot capture, with the Random Forest reducing average error by approximately 40%.

The default Random Forest is selected as the best model. It achieves the highest R^2, the lowest RMSE, and the lowest Maximum Error (€84.5M). Its Median Absolute Error of €123,670 means that for more than half of all players, the prediction is off by less than €124,000. Although the tuned Random Forest has marginally better MAE, its higher RMSE and lower R^2 suggest slight overfitting to cross-validation folds. The simpler configuration offers a better bias-variance trade-off. The Neural Network is competitive with the best

MAPE at 81.3%, but trails in R^2 and RMSE while requiring additional preprocessing such as log-transforming the target.

Feature importance analysis reveals that the strongest predictors are age-related features, performance metrics (goals per 90, goal contributions), league quality, and contract duration — aligning with football domain knowledge. By comparing predicted values with actual market prices, the model identifies discrepancies indicating potential undervaluation or overvaluation, providing actionable insights for transfer decisions.

Several limitations should be noted. MAPE remains high (81–95%) across all models, driven by low-value players where small absolute errors translate to large percentage errors; for the target use case of players valued above €1 million, the effective MAPE is substantially lower. Maximum Error reaches €84–95 million, indicating difficulty with elite players whose values are driven by brand power and bidding wars. Temporal dynamics such as market inflation and post-tournament hype are not captured.

5. Conclusion

This project demonstrates a complete Machine Learning pipeline applied to football player valuation. The Random Forest model was selected as the best performer, explaining 73.7% of market value variance with a mean absolute error of €773,568 and a median error of €123,670. Key technical contributions include position-aware career phase engineering, a five-tier league quality encoding, and comprehensive benchmarking of seven models across six metrics.

From a business perspective, the model provides mid-tier clubs with a data-driven tool to screen thousands of players and flag undervalued talent, reducing the financial risk of transfer decisions. Future work should explore gradient boosting methods (XGBoost, LightGBM), advanced tracking data (expected goals, progressive carries), temporal market features, and deployment as an interactive web application

References

Transfermarkt. Player data, market valuations, and injury records. https://www.transfermarkt.com

Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Model Evaluation Dashboard